

PUNEETH MUTHINENI

Software Engineer | AI & Full-Stack

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SUMMARY

Software engineer with close to two years of experience building and shipping production backend and AI-powered features end to end, plus an MS in Computer Science. Works across the full stack: React interfaces, Node.js service gateways, and Python/FastAPI backends behind REST/JSON APIs, backed by PostgreSQL and MongoDB. Builds agentic workflows with modern LLM APIs (Claude/OpenAI) and orchestration libraries (LangGraph, LangChain), applying prompt design, tool calling, structured outputs, and deterministic guardrails so AI-assisted behavior stays predictable and testable. Owns the surrounding infrastructure: data pipelines, pytest evaluation harnesses, MLflow experiment tracking, Docker, CI/CD, and observability. Ships in small fast increments and works directly with stakeholders to clarify ambiguous requirements and iterate.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript (Node.js / React)

Generative AI & LLMs: Claude/OpenAI APIs, prompt design, structured outputs, tool & function calling, grounding, hallucination mitigation, output validation, guardrails

Agentic AI & Orchestration: LangGraph, LangChain, agentic workflows, multi-step reasoning, tool/API calling, deterministic fallback behavior

Backend & APIs: FastAPI, Flask, Node.js, REST/JSON APIs, WebSockets, microservices, Pydantic validation, service boundaries, API contracts

Frontend: React, component-based UI, front-end to back-end integration

Databases & Data Engineering: PostgreSQL, MongoDB, SQL, pandas, NumPy, data cleaning/transformation/validation, JSON processing, data modeling

Cloud & DevOps: AWS (S3, EC2), Azure, GCP, Docker, Linux, GitHub Actions, Jenkins, CI/CD pipelines

MLOps / LLMops & Observability: MLflow tracking, pytest evaluation harnesses, regression testing, retrieval/model evaluation, monitoring, logging, alerting, production troubleshooting

RAG & Vector Search: Document ingestion, chunking, embeddings, FAISS vector indexing, semantic retrieval, retrieval evaluation

Machine Learning & NLP: scikit-learn, PyTorch, Hugging Face Transformers, XGBoost, BERT, hyperparameter optimization (Optuna), model evaluation

WORK EXPERIENCE

Acentle

Jan 2025 – Present

Software Developer (Freelance / Contract)

- Built, shipped, and supported backend and AI-enabled services across 5+ production releases, owning design, implementation, testing, deployment, and post-release support with minimal supervision.
- Developed SyncDev Spatial, a Linux-based Python platform exposing 10+ REST/JSON API endpoints with WebSocket-based real-time communication, Docker deployment, and monitoring/logging, reducing manual operational overhead by ~40%.
- Integrated backend services with REST APIs, JSON payloads, and SQL-backed data logic, adding monitoring workflows that improved data availability and reliability.
- Designed Python and SQL data flows to clean, transform, validate, and serve operational data for downstream workflows, reporting, and automation.
- Automated build, lint, and pytest workflows in CI/CD pipelines, improving release consistency and cutting deployment-validation time by ~30%.
- Diagnosed production incidents by tracing logs, alerts, and API behavior to root cause, then validated fixes and hardened reliability; applied input/output validation, edge-case testing, and code review before every release.

Nice Interactive

Jan 2024 – Jun 2024

Software Developer Intern

- Built Python backend features and REST API integrations supporting data exchange, service functionality, and internal application workflows.
- Wrote SQL across 3+ application modules for data retrieval, reporting, backend workflows, and application-level data validation.
- Reproduced, debugged, and resolved defects; added unit, integration, and regression tests to improve reliability and reduce repeat defects.
- Shipped through Git-based workflows (feature branches, pull requests, code review, CI/CD), incorporating senior-engineer feedback on readability and maintainability.
- Collaborated with developers and stakeholders in an agile environment to clarify ambiguous requirements and deliver backend updates on time.

PROJECTS

CareCompass — Full-Stack AI Product with Agentic Workflow

React · Node.js · Python/FastAPI · LangGraph · MongoDB · Docker · pytest

- Shipped a production-style full-stack app end to end — React UI, Node.js service gateway, Python/FastAPI backend with REST APIs and request validation — owning complete features from interface through persistence.
- Integrated a LangGraph agentic workflow into the live product path for AI-assisted recommendations, while the service layer stayed responsible for validation, persistence, and logic.
- Engineered deterministic safety guardrails and fallback paths so agent outputs stayed predictable, testable, and monitorable; Dockerized with pytest + CI gates on every change.

Unisphere — Enterprise RAG & LLM Workflow Platform

Python · FastAPI · AWS · LangChain · LangGraph · FAISS

- Designed and deployed an enterprise RAG and semantic-search pipeline on AWS — ingestion, chunking, embeddings, FAISS indexing, semantic retrieval, and grounded LLM responses over multi-source data.
- Built LangChain/LangGraph orchestration with prompt design, tool/API calling, structured outputs, and intermediate reasoning, exposed through FastAPI REST/JSON endpoints.
- Established a retrieval evaluation loop (correctness, latency, answer quality, consistency) to tune chunking, prompts, and retrieval behavior ahead of production.

MLOps Prediction Service — Model Training, Tuning & Evaluation Harness

Python · scikit-learn · FastAPI · PostgreSQL · MLflow · Optuna · pytest

- Deployed a trained model as a FastAPI + PostgreSQL inference service with Pydantic validation, structured prediction responses, and SQL-backed data models.
- Ran Optuna hyperparameter optimization across 96 tracked MLflow experiments, improving F1 by 3.3% and AUC by 2.9%; built a 14-test pytest CI evaluation harness to gate quality before deployment.
- Engineered reproducible Python data pipelines for cleaning, transformation, validation, and feature preparation, with monitoring, logging, and experiment documentation.

EDUCATION

M.S. Computer Science, University of Massachusetts Boston	GPA 3.6 / 4.0
B.Tech Computer Science & Engineering, VIT-AP University	CGPA 8.2 / 10

PUBLICATIONS & CERTIFICATIONS

BERT-QA: Question Answering with NLP & Entity Recognition	Published 2024
Google Cloud Professional Machine Learning Engineer	Google Cloud
AWS Certified Machine Learning Engineer – Associate	Amazon Web Services
Google AI Essentials	Google